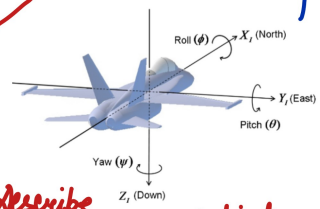
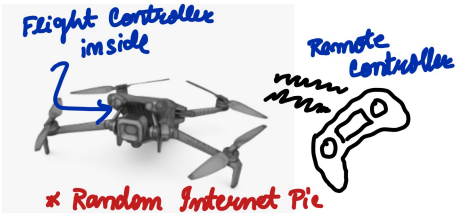
Drone Control Loop

IMU → Accelerometer, Gyroscope → Sensor Fusion Algorithm → Euler Angles → Yaw, Pitch, Roll → PID Controller → Flight Controller

* Throttle: The current, real-time number the Remote Controller (RC) sends the Flight Controller (FC), comes from joystick.
Let's say range from 0 (motor off) to 1000 (full power)



Describe 3D orientation of object with respect to fixed reference frame



* Random Internet Pic

Example:

Drone Flying → Wind gust → Unstable! → PID
To stabilize: Push left up, Drop right down
→ Stable!

Left motors spin faster, Right slower
So motor_left = throttle + roll correction
motor_right = throttle − roll correction
} Assuming all motors spin in the same direction

Motor PWM

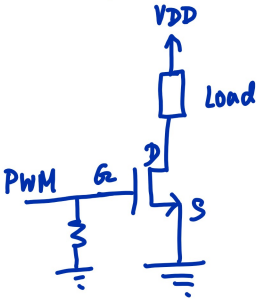
① Using Electronic Speed Controllers (ESCs) → expect a pulse every 20ms.

* Pulse of 1ms = 0% Throttle
* Pulse of 2ms = 100% Throttle

External ESC → may add weight!

So, take throttle (0 - 1000) → map to pulse width (1ms - 2ms) → Set PWM pulse

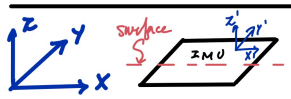
② Simple low side switching (+ flyback diode connected across motor load (inductive) to prevent damage from voltage spikes)



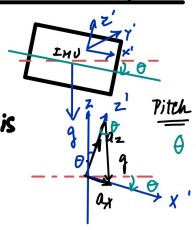
So, can map motor_left & motor_right directly to PWM percentage

* Note: Choose high PWM frequency to reduce vibration (≈ 20 kHz)

Calculate Roll and Pitch angles



Roll: Rotation around X-axis
Pitch: " Y-axis
Yaw: " Z-axis



$$\theta = \tan^{-1}\left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}}\right)$$

Similarly, Roll:

$$\phi = \tan^{-1}\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right)$$

Remote Controller

Inputs


Joystick1


Joystick2

Sample ADC @ 100 Hz
12 bit resolution - 0 to 4095

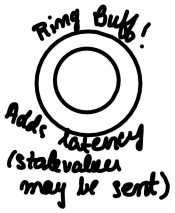
analog_joystick.h
analog_joystick.c
1. Read ADC (4 channels)

↓ Calls

decode_joystick.h
decode_joystick.c
2. Decode ADC readings (Mode 2 config; Desired Throttle, Yaw, Pitch, Roll) — TYPR

↓ Gives Semaphore to wake ESB

radio_esb.h
radio_esb.c
3. Prepare & TX Data Packet



To prevent data being Read as it's being written to, use mutex:

Joystick → Lock → Write → Unlock → Give Sem → ESB

Flight Controller

IMU → Sensor fusion → PID → Motor PWM

ESB Rx Interrupt